The growth-needs clinic

Teacher guide and worked answers

Identify requirements for healthy plant growth and use height plus leaf evidence to evaluate a care claim.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsb/inputs/Build SOW 2026-2027.xlsx | BUILD Weekly - Summer | C32

Identify what plants need to grow well (light, water, warmth, nutrients).

Chronological continuation of the supplied SOW. This lesson develops or reviews the named outcome; it does not claim an accreditation award.

Prior learning

Roots absorb water and mineral nutrients.

Green leaves make sugars using light.

Success evidence

Name light, water, suitable warmth and minerals.

Include air and sufficient space in a care plan.

Use leaf condition as well as height.

Equipment

Printed shared evidence and one chosen pupil route; pencil; optional ruler; projector or offline HTML.

Preparation

Print the shared evidence, one response route and exit page. Routes are alternatives, not a workload ladder.

Read the worked answers. Prepare an oral, pointing or adult-scribed route where useful.

Practical care

This is a paper evidence clinic; no plants are deliberately deprived or overheated.

For optional care observation, use adult-selected non-toxic plants, no tasting, and wash hands afterwards.

Ready to teach alternative

All core reasoning uses the supplied evidence and can be completed in 40 minutes without a live practical.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Which plant needs a better care plan?
2	Retrieval	2-5	Retrieve the inputs

Slide	Stage	Minutes	Focus
3	I do	5-12	I check the whole set of needs
4	I do	Within stage	I refuse to use height alone
5	I do	Within stage	I separate seed and green-plant stages
6	We do	12-16	The evidence clinic: compare A and B
7	We do	16-20	Choose a repair that fits the condition
8	Hinge	20-23	B is 13 cm and pale. A is 8 cm and green.
9	You do	23-25	Choose your science route
10	You do	25-35	Make your own evidence record
11	You do	Within stage	Check what the evidence can show
12	Review	35-38	Lundy loop: improve the shared account
13	Review	Within stage	Keep the useful change
14	Exit	38-40	Three final checks

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Which plant needs a better care plan?

Introduce constructed records of matched young bean seedlings. Do not present this as the class's actual experiment.

2 Retrieval Retrieve the inputs

1 roots. 2 light, water and carbon dioxide from air; light enough for the short prompt. 3 no.

3 I do I check the whole set of needs

Think aloud: "Providing water does not make light unnecessary. I check each requirement. I ask what is suitable for this plant, rather than assuming more is always better."

4 I do I refuse to use height alone

Shares I do time. Think aloud: "13 is greater than 8, but pale small leaves change my judgement. Dark-grown seedlings can become elongated; taller is not a general health score." The examples show selected seedlings from the constructed groups.

5 I do I separate seed and green-plant stages

Shares I do time. Think aloud: "A seed starting in darkness does not prove that an established green plant can grow well indefinitely without light." Seeds also differ in light requirements.

6 We do The evidence clinic: compare A and B

W1: B is taller but pale; A shows stronger evidence of healthy growth here. All seedlings started about 3 cm. Two plants per group are supplied; not enough for a universal conclusion.

7 We do Choose a repair that fits the condition

W2: C adjust watering appropriately; D provide suitable warmth. Do not tell pupils to flood a pot or put a plant on a radiator. The cases stipulate the changed condition, not a general diagnosis from drooping alone.

Reveal: Address the stated missing requirement; keep other needs suitable.

8 Hinge B is 13 cm and pale. A is 8 cm and green.

Correct second option. Return to leaf condition if height alone dominates.

A. B must be healthier because it is taller. - Pale small leaves challenge that conclusion.

B. Height alone cannot establish which grows well. - Leaf condition and suitable inputs also matter.

C. No green plant needs light. - Sustained photosynthesis needs light energy.

9 You do Choose your science route

Set ONE route. Use the shared constructed table. Do not diagnose an actual plant from one symptom. Accept suitable care language and more than one observation; do not reward a universal care recipe.

10 You do Make your own evidence record

Use the shared constructed table. Do not diagnose an actual plant from one symptom. Accept suitable care language and more than one observation; do not reward a universal care recipe.

11 You do Check what the evidence can show

Shares independent time. Ask for a reason, not just a correct word. Use the shared constructed table. Do not diagnose an actual plant from one symptom. Accept suitable care language and more than one observation; do not reward a universal care recipe.

12 Review Lundy loop: improve the shared account

Listen to a selected pupil revision and visibly adopt a scientifically accurate contribution. Offer private, written or partner feedback.

Reveal: A taller pale seedling does not automatically show healthier growth.

13 Review Keep the useful change

Shares review time. Name how a pupil contribution changed a label, method or conclusion. Do not rank pupils.

Reveal: A taller pale seedling does not automatically show healthier growth.

14 Exit Three final checks

Collect brief individual evidence. Use the answer key and re-teach the specified misconception if needed.

Answers and assessment

Retrieval 1-3

1 roots. 2 light, water and carbon dioxide from air. 3 no.

We do W1-W2

W1 B is taller (12-13 cm versus 8-9 cm); pale small leaves matter, while A is green with several leaves. W2 adjust C’s water supply appropriately; move D to suitable warmth while preserving other needs.

Hinge

Second option correct. Greater height is only one observation; green plants need light for sustained photosynthesis.

S1-S4

S1 any four from light, water, suitable warmth, minerals, air and space; ensure the four named SOW requirements are taught. S2 no. S3 pale small leaves. S4 water.

M1-M4

M1 B taller but pale/small leaves; A shorter and green with several leaves. M2 appropriate watering because C explicitly had too little; avoid flooding. M3 excessive heat/water can be unsuitable and species differ. M4 minerals are needed substances, whereas light supplies energy for photosynthesis.

T1-T4

T1 B 12-13 cm but pale and small-leaved; A 8-9 cm and green. T2 small sample, one species, one stage and one interval cannot justify all plants. T3 stored seed reserves can support initial growth; sustained green growth needs photosynthesis, and seeds differ in light requirements. T4 e.g. in these examples, dark-grown seedlings were taller but paler; height alone did not show healthy growth.

Exit E1-E3

E1 any two valid requirements. E2 no, include leaf condition and other evidence. E3 no: different requirements have different roles.

R1-R2

Responses vary. Credit a scientific revision, evidence-based defence or relevant next question. A private or oral response has equal value.

Teaching and assessment notes

40 minutes: opening 2, retrieval 3, I do 7, We do 8, hinge 3, independent 12, review 3, exit 2. Zero-minute slides share the named stage time.

BUILD uses the supplied Year 3 science anchor with age-respectful contexts for older pupils. Supported, standard and stretch are selectable routes within the pathway.

Interactive We do: predict first, test against the controlled SVG model or supplied evidence, then explain or revise. Shared W tasks offer the equivalent paper discussion. Animation is a teaching representation, not filmed experimental evidence.

Keep constructed evidence separate from actual class observations. Do not invent measurements or require internet research.

Secure core includes light, water, appropriate warmth and minerals plus awareness of air/space. A pupil should reject height-only judgement using the pale-leaf clue.

Access and responsive teaching

Supported

Short choices, word bank and pointing or adult-scribed reasons; same core scientific goal.

Standard

Make a short evidence-based explanation or record using the supplied information.

Stretch

Apply the core idea to a changed case, evaluate evidence or identify a limit.

Regulation

Preview the sequence. Offer seated observation, quiet paired rehearsal, private feedback and a pass from public sharing. Routes respond to current access needs, not a diagnosis.

Misconceptions to check

The tallest seedling must be healthiest.

A dark-grown seedling can elongate and become pale; height alone is insufficient.

Check: Compare groups A and B using leaf condition.

More water and warmth always help.

Too much water or unsuitable heat can harm growth; suitable amounts depend on the plant.

Check: Ask for appropriate, not maximum, provision.

Every seed needs exactly the same light to germinate.

Species differ; many use stored reserves before green growth is established.

Check: Distinguish germination from sustained green growth.

Word	Meaning
requirement	Something needed for a process or life.
mineral nutrient	A substance needed for healthy growth.
germination	A seed beginning to grow.
evidence	Information used to judge a claim.

Scientific references

Department for Education: science programmes of study

Checked 8 September 2026. Year 3 plants, rocks, light and forces underpin the SOW. Accessible teaching models are adapted to the named lesson objective.

Science and Plants for Schools: plant biology animations

Checked 8 September 2026. Scientific background for our original simplified diagrams: photosynthesis, growth and transport. No external animation is required or embedded.

Royal Horticultural Society: sowing seeds indoors

Checked 8 September 2026. Germination uses stored reserves and requires appropriate water, oxygen and temperature; seedling care differs from starting germination.

Royal Horticultural Society: germination guide

Checked 8 September 2026. Species differ in their germination requirements, including whether light is required.